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MILITARY SPECIFICATION

PROPELLANT, M31(T34), FOR USE IN CHARGE PROPELLING M45, FOR PROJECTILES M365 AND M357, FOR 120MM GUN, M58

1. SCOPE

1.1 This specification covers the M31(T34) propellant for one type of propelling charge designated as M45 for 120MM Gun, M58.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for

MIL-I-45208 — Inspection Requirements; General Specification for

STANDARDS

MIL-STD-105D — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-414 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-652 — Solid Propellant for Cannon Requirements and Packing

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The propellant shall be M31(T34), Type II, and conforming to the requirement of Standard MIL-STD-652.

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3.2 Ballistics. The propellant, when fired in the weapon and with the weight of projectile for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.1.

3.2.1 *Velocity and pressure.* At 70°F., a weight of propellant assessed as required to impart a service velocity of 762 meters per second (mps) to the projectile shall not produce a maximum rate pressure of more than 18,000 pounds per square inch, Piezo equivalent pressure.

3.2.2 *Uniformity of velocity.* The velocity shall be not greater than 770 mps nor less than 754 mps. The standard deviation in velocity for the lot, at 70 degrees Fahrenheit, shall not exceed 1.3 mps.

3.3 Workmanship. The best commercial practices shall be used in the formulation of the propellant furnished under this specification and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply:

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of analysis on all material produced directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

(h) Eight copies of propellant description sheet, 00 Form 1204, containing the applicable data.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the Agent's authority will not be required with subsequent certificates unless during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of one or more batches of M31 (T34) Propellant produced by one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of M31 (T34) Propellant which has been subjected to the same unit chemical or physical mixing process intended to make the final product homogenous. Product shall be submitted for inspection in accordance with Standard MIL–STD–105D (or Standard MIL–STD–1235 when applicable).

4.2.2 *Testing.*

4.2.2.1 Ballistics (sec 3.2) major defect. A sample of 1,400 pounds shall be randomly selected from the lot for the ballistic test.

4.2.2.1.1 *Velocity.* A sample of 10 propelling charges shall be assembled for this test. If the mean plus the product of 1.65 times the standard deviation (see 6.2) exceeds the upper limit specified, or if the mean minus the product of 1.65 times the standard deviation is less than the lower limit specified or if the sample standard deviation times 0.73 exceeds the limit specified the lot shall be rejected. The test shall be performed as specified in 4.3.1 Code No. 01001.

4.2.2.2 *Retest.* If for any reason the proof director considers that the test conditions have detrimentally affected the rest results, he shall declare the results invalid and perform a new test on additional charges as required.

4.3 Test methods and procedures.

4.3.1 *Ballistics.* These tests shall be conducted at a Government Proving Ground in accordance with the applicable Ordnance Proof Manual and Acceptance Test Procedure. A ten-round uniformity series shall be fired and the velocity mean and standard deviation shall be calculated after reducing results to standard conditions and the pressure shall be reported as Piezo Equivalent pressure.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. The propellant shall be packed and marked in accordance with Standard MIL–STD–652.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) *Process.* Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture. Any deviation from this manufacturing process must be submitted in writing to the procuring activity prior to being put into effect.

6.2 Standard deviation. The standard deviation will be calculated with (N–1) as the devisor in a standard statistical technique to

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that shown on Standard MIL–STD–414, Section B; example B–1.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement action, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulted, furnished, or in any way supplied the said drawings, specifications, or other data is not intended to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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